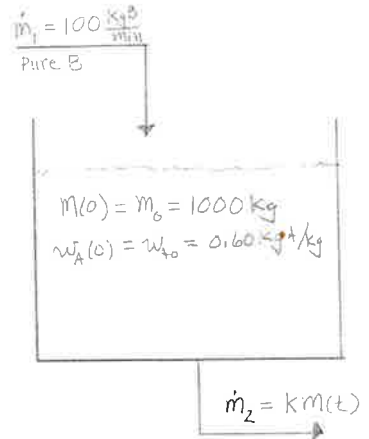


To Receive Full Credit, You Must Show All Your Work

Initially at $t = 0$, a mixing tank contains $m(0) = 1000$ kg of a binary liquid mixture of species A and B. The initial mass fraction of species A is $w_A(0) = 0.60$. The inlet flow rate is $\dot{m}_1 = 100$ kgB/min of pure B. The outlet flow rate is given by $\dot{m}_2 = k \cdot m(t)$ where $k = 2.0 \text{ min}^{-1}$ and $m(t)$ is the mass in the tank as a function of time. Assume the tank is well mixed throughout the process.

- (a) At steady state, how much mass, in kg, is left in the tank? That is, when $t \rightarrow \infty$, what is $m(t \rightarrow \infty)$?
- (b) Derive a general expression for $m(t)$. How long does it take for $m(t)$ to drop to 100 kg?
- (c) Derive a general expression for $m_A(t)$, the mass of A in the tank. At the time calculated in part (b), what is w_A , the mass fraction of A in the tank?



$$t = 0,$$

$$\dot{m}_1 = 100 \text{ kgB/min}$$

$$\dot{m}_2 = k \cdot m(t)$$

$$m(0) = 1000 \text{ kg}$$

$$w_A(0) = .60$$

$$w_B(0) = .40$$

$$\frac{dm_{\text{sys}}}{dt} = \sum \dot{m}_i - \sum \dot{m}_j$$

$$\frac{dm_{\text{sys}}}{dt} = 100 \frac{\text{kg}}{\text{min}} - 2 \cdot m(t) \frac{\text{kg}}{\text{min}}$$

$$\int_0^{\infty} 100 - 2m(t) dt$$

$$100t - \frac{2m(t)^2}{2} \Big|_0^{\infty}$$

$$\int dm_{\text{sys}} = \int (100 - 2m(t)) dt$$

$$100t_0 - m(t_0)^2 - 0$$

$$t_0(100 - m(t_0))$$

